Datasets

All sources below were accessed programmatically; nothing was digitised by hand.
Access dates: 2026-09-29/30.

Glacier inventories (GLIMS WFS, `https://www.glims.org/geoserver/ows`)

Layer	Use	Chhota Shigri record
`GLIMS:RGI_SouthAsiaWest` (RGI 6.0)	ID cross-reference	`RGI60-14.15990`, GLIMS `G077508E32224N`, 16.764 km², source date 2002-08-02, zmin 4280 m, zmax 5764 m, Lmax 8600 m
`GLIMS:RGI2000-v7.0-G-14_south_asia_west_epsg3857` (RGI 7.0)	ROI, analysis domain, terminus attribute, neighbour divides	`RGI2000-v7.0-G-14-30579`, GLIMS `G077513E32227N`, 16.772 km², 2002-08-02
`GLIMS:GLIMS_Glacier_Outlines`	historical context / reference	7 analyses (table below)

GLIMS outlines of Chhota Shigri (area recomputed in UTM 43N):

GLIMS analysis	Date	Institution	Area (km²)
1065541	"1450" (LIA reconstruction)	Univ. Leeds	49.13 (glacier complex)
1127326	2000-10-15	KOPRI	13.01
860910 / 1083522	2002-08-02	Nagoya Univ. (GAMDAM) / Univ. Colorado (RGI)	16.77
119262	2002-08-02	Univ. Zurich	14.87
60228	2002-09-28	ICIMOD	16.14
154108	2009-10-25	ICIMOD	11.65

Important: these outlines from the same 2000–2009 period differ by up to 5 km² (±30 %). The spread is dominated by different ice-divide / tributary partitioning, not by ice loss. All area statistics in this project are therefore computed inside one fixed *analysis domain* (RGI7 outline + 1 km buffer, minus the 33 neighbouring RGI7 glaciers), so that only margin changes affect the time series.

Coordinate note: the frequently quoted "32.28° N, 77.58° E" for Chhota Shigri coincides with the RGI7 terminus of *Bara Shigri* (32.274° N, 77.585° E). The RGI7 centroid of Chhota Shigri is 32.224° N, 77.508° E (RGI6 centroid 32.224° N, 77.508° E). The ROI is derived from the
inventory polygon, never from a point buffer.

RGI7 centerlines (`RGI2000-v7.0-L`) are distributed only by NSIDC behind a NASA Earthdata login (HTTP 401 without credentials). A centerline is instead generated automatically (`src/change_detection/centerline.py`). If an Earthdata account becomes available, the official
product can replace it.

Sentinel-2

- **Discovery and screening:** Microsoft Planetary Computer STAC, collection `sentinel-2-l2a`
  (scene metadata + Scene Classification Layer, SCL).
- **Model input (TOA, L1C):** Planetary Computer serves only L2A. GlaViTU was trained on L1C
  top-of-atmosphere reflectance (`utils/dataset_stats.pickle` : optical min ≈ -0.027, max ≈ 2.7,
  i.e. harmonised TOA reflectance). L1C granules are read from the public Google Cloud bucket
  `gcp-public-data-sentinel-2` (anonymous HTTPS, windowed JPEG2000 reads over the ROI only).
- **Harmonisation:** reflectance = (DN + RADIO_ADD_OFFSET) / QUANTIFICATION_VALUE from each
  product's `MTD_MSIL1C.xml` (offset −1000 for processing baseline ≥ 04.00, 0 before), equivalent
  to `COPERNICUS/S2_HARMONIZED` .
- **Tile:** MGRS 43SGR, relative orbit 105 (covers the whole ROI).
- **Cloud masking:** s2cloudless (the algorithm behind the "Sentinel-2 Cloud Probability" product), run on the L1C bands at 20 m; threshold 0.4, 3 px dilation. SCL cloud/shadow is used

only for fast pre-screening, because SCL frequently labels bright ice/snow as cloud over Chhota Shigri (e.g. 2019-09-17: SCL 11 % vs s2cloudless 2.3 % over the domain).
- Scene IDs, dates, radiometric offsets and per-candidate statistics are written to `data/sentinel2/selection_<year>.csv` and `data/sentinel2/<year>/scene.json`.

DEM

Copernicus DEM GLO-30 (`cop-dem-glo-30` on Planetary Computer), tile `Copernicus_DSM_COG_10_N32_00_E077_00_DEM`; bilinear to the 10 m grid; slope/aspect with Horn's
method at 30 m, then upsampled.

GlaViTU benchmark (Maslov et al. 2025)

NIRD archive doi:10.11582/2024.00168 (CC-BY-4.0). The full tile-based dataset is 411 GB (`20230905_{train,val,test}_global_ps384.hdf5` = 243 / 81 / 86 GB); the Google Drive "demo" copy
is 38 GB. Tile counts (full dataset):

Split	Total	HMA
train	1636	165
val	547	55
test	556	57

NIRD supports HTTP byte ranges, so only the HMA tiles are read (`src/data/benchmark.py`) and
stored locally as float32 (`data/benchmark/hma_{train,val,test}.hdf5`). The official split is kept.

Pretrained weights

Google Drive folder linked from the GlaViTU README (30 checkpoints). Used: `glavitu_global_weights.h5` (M0), `glavitu_finetuning_HMA_weights.h5` (official global→HMA
fine-tuning, "M1-official"), `glavitu_regional_HMA_weights.h5` (HMA-only training). SHA-256 in
`models/pretrained/SHA256SUMS`.